

Lecture 26.

I Independent r.v.

II Algebra with r.v.

III Variance

Defn: Random variables $X, Y: S \rightarrow \mathbb{R}$ are independent if $\forall r_1, r_2 \in \mathbb{R}$ the events $\{X=r_1\}$ and $\{Y=r_2\}$ are independent.

ex:

$$X = \begin{cases} 1 & \text{with prob } q \\ 0 & \text{" " } 1-q. \end{cases}$$

Bernoulli(q)

distribution of X . : $P(\{X=r\})$ for every $r \in \mathbb{R}$.

$$\mathbb{E} X = 1 \cdot P(X=1) + 0 \cdot P(X=0) = q.$$

ex 2:

Let $X_1, X_2, \dots, X_n$ be independent Bernoulli(q) random variables. → coin flip w/ bias q .

Let $X = X_1 + X_2 + \dots + X_n$. ← # of heads

Last lec:

$$P(X=r) = \binom{n}{r} q^r (1-q)^{n-r}$$

Binomial(n, q)
distribution.

$$EX = \overset{\text{linearity}}{nq}.$$

< Demo: $q=0.5$, $EX = n/2$ >.

II Algebra with r.v.

Already saw: $X+Y$: $\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y$
 $cX, c \in \mathbb{R}$: $\mathbb{E}(cX) = c \mathbb{E}X.$

Def: If $X, Y: S \rightarrow \mathbb{R}$, let
 $XY: S \rightarrow \mathbb{R}$ by $(XY)(s) = X(s)Y(s)$
 $\forall s \in S.$

ex3: If X_1, X_2 are indep. Bernoulli(q),

$$X_1 X_2 = \begin{cases} 1 & \text{if } X_1=1 \text{ and } X_2=1 \\ 0 & \text{otherwise.} \end{cases}$$

q^2
2 heads
 $1-q^2$

$$\mathbb{E} X_1 X_2 = 1 \cdot \mathbb{P}(X_1=1 \text{ and } X_2=1) \stackrel{\text{indep}}{=} \mathbb{P}(X_1=1) \mathbb{P}(X_2=1) = q \cdot q = \underline{\underline{q^2}}.$$

sum $\mathbb{E}X_1 \mathbb{E}X_2 = q \cdot q = q^2$

ex4: X, Y independent dice rolls.

$XY =$ product of the two numbers.

Theorem: If $X, Y: S \rightarrow \mathbb{R}$ are independent random variables, then

$$\mathbb{E} XY = \mathbb{E} X \mathbb{E} Y.$$

<Algebra, Skip>.

Q11: Is $\mathbb{E} X^2 = (\mathbb{E} X)^2$ for all X ?

If X and X are indep: $\mathbb{E} XX = (\mathbb{E} X)^2$

$\forall r_1, r_2 : \{X=r_1\} \{X=r_2\}$ are indep.

$$P(\{X=r_1\} \cap \{X=r_2\}) = P(X=r_1)P(X=r_2).$$

$r_1=r_2=r :$ $P(\cancel{X=r}) = P(X=r) \cancel{P(X=r)}$

$P(X=r) = 0 \text{ or } 1.$ Not true in general.

$\therefore X$ and X are not independent.

Counterexample: $X \sim \text{Bernoulli}(q)$.

$EX = q.$ $X^2 = \begin{cases} 1^2 & \text{with prob. } q \\ 0^2 & \text{" } 1-q. \end{cases}$

$EX^2 = 1^2 \cdot P(X^2=1^2) + 0^2 \cdot \cancel{P(X^2=0^2)}$
 $= q //$

$$\mathbb{E}X^2 = q \neq (\mathbb{E}X)^2 = q^2 \text{ unless } q=0 \text{ or } 1.$$

Special Case: If $X = c$ for some $c \in \mathbb{R}$
 (Scalar / Deterministic), then ① $\mathbb{E}X^2 = (\mathbb{E}X)^2$
 ② X is independent of X .

More generally: Can consider expressions like

$$(X+Y)^2(s) = X^2(s) + 2X(s)Y(s) + Y^2(s).$$

Remark: $\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y$ in general,
 regardless of independence.
 but $\mathbb{E}XY \neq \mathbb{E}X \mathbb{E}Y$ in general.

ex 5:

$E, F \subseteq S$ events.

$\downarrow$

$1_E, 1_F$ indicators.

$$1_E 1_F^{(s)} = \begin{cases} 1 & \text{if } s \in E \text{ and } s \in F \\ 0 & \text{otherwise} \end{cases}$$

$$= 1_{E \cap F}^{(s)}.$$

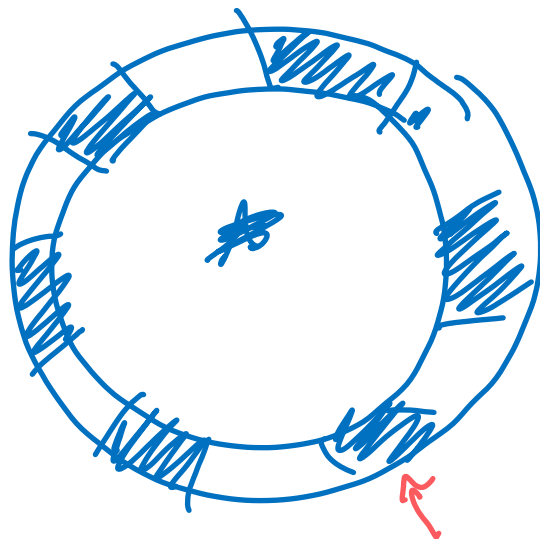
$$\text{So } \mathbb{E} 1_E 1_F = \mathbb{E} 1_{E \cap F} = P(E \cap F)$$

$\neq P(E) P(F)$
in general.

III Variance

Motivation: Given X , how close is X typically to $\mathbb{E}X$?

ex 6: Roulette:



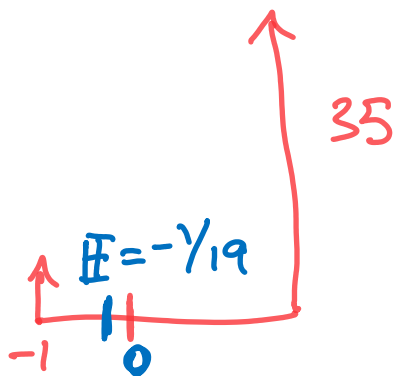
38 numbers
18 red
18 black
2 green.

Bet types:

① Single Number: 35:1 odds: win: \$35
lose: \$1.

X = Amount of money won after 1 trial.

$$\mathbb{E}X = \left(\frac{1}{38}\right)(35) + \left(\frac{37}{38}\right)(-1) = \frac{-2}{38} = -\frac{1}{19}$$

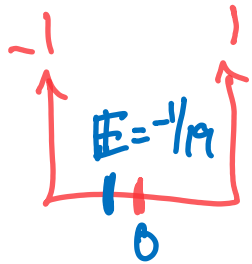


$$V(X) = E(X + 1/19)^2 = \left(35 + \frac{1}{19}\right)^2 \left(\frac{1}{38}\right) + \left(-1 + \frac{1}{19}\right)^2 \left(\frac{37}{38}\right)$$

$\bar{u}_x \sim \underline{\underline{33.2}} \text{ high}$

② Bet on Red: 1:1 odds. win \$1, red
lose \$1, black.

Y = amount of money won after 1 trial.



$$EY = \left(\frac{18}{38}\right)(1) + \left(\frac{20}{38}\right)(-1) = -\frac{1}{19} //$$

$$V(Y) = \left(1 + \frac{1}{19}\right)^2 \left(\frac{18}{38}\right) + \left(-1 + \frac{1}{19}\right)^2 \left(\frac{20}{38}\right)$$

~ 1 low variance.

Defn: If a random variable has $\mathbb{E}X = \mu_X$
the variance of X is $V(X) = \mathbb{E}(X - \mu_X)^2$.

Remark: The standard deviation
 $\text{stdev}(X) = \sqrt{V(X)}$ measures
the "typical" deviation of X from $\mathbb{E}X$.
($\hookrightarrow$ heavily influenced by larger deviations.)

Remark: $V(X) = 0 \iff \mathbb{E} \overbrace{(X - \mu_X)^2}^{=0} = 0$
 $\iff P(X - \mu_X = 0) = 1$.
 $\iff X = \mu_X$ always equal to $\mathbb{E}X$.

Remark: Why not the following?

$$\mathbb{E}(X - \underbrace{\mu_X}_{\text{scalar}}) = \mathbb{E}X - \mathbb{E}\mu_X = \mu_X - \mu_X \stackrel{\text{always zero.}}{=} 0.$$

$$\mathbb{E}|X - \mu_X| = \text{pain to calculate.}$$

$$\star \mathbb{E}(X - \mu_X)^2 = \langle X - \mu_X, X - \mu_X \rangle$$

in an inner product space
(Math 54).

(Q) Relation between $\mathbb{E}|X - \mu_X|$ and $\sqrt{V(X)}$?

$$(A) \quad \mathbb{E}|X - \mu_X| \leq \sqrt{V(X)}. \quad (\text{HW, extra cred}).$$

Remark:

$$\begin{aligned} V(X) &= \mathbb{E}(X - \mu_X)^2 \\ &= \mathbb{E}(X^2 - 2\mu_X X + \mu_X^2) \\ &\stackrel{(\text{lin.})}{=} \mathbb{E}X^2 - 2\mu_X \underbrace{\mathbb{E}X}_{\mu_X} + \mu_X^2 \\ &= \mathbb{E}X^2 - 2\mu_X^2 + \mu_X^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2. \end{aligned}$$

cf. poll.

ex 7: $X \sim \text{Bernoulli}(q)$ $q \in [0, 1]$,

$$V(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = q - q^2 = \underline{\underline{q(1-q)}}.$$

Theorem : If X and Y are independent random variables,
then $V(X+Y) = V(X) + V(Y)$.

Proof: Let $\mu_x = EX$, $\mu_y = EY$. By linearity,

$$E(X+Y) = EX + EY = \mu_x + \mu_y.$$

$$V(X+Y) = E(X+Y)^2 - (\mu_x + \mu_y)^2$$

$$\stackrel{\text{expand}}{=} E(X^2 + 2XY + Y^2) - (\mu_x^2 + 2\mu_x\mu_y + \mu_y^2)$$

$$\stackrel{\text{lin.}}{=} (EX^2 - \mu_x^2) + (EY^2 - \mu_y^2) + \underline{\underline{(2EXY - 2\mu_x\mu_y)}}$$

$EXY = EXEY$
by independence

$$= V(X) + V(Y) + 2 \cancel{\mathbb{E}X\mathbb{E}Y} - 2 \mathbb{E}X\mathbb{E}Y \xrightarrow{0}$$

($\mathbb{E}Y = \mathbb{E}X$) $\square$

Corollary: If $X_1, \dots, X_n$ are pairwise independent
 (i.e. $\forall i \neq j, X_i$ and X_j are independent) random
 variables, then $V(X_1 + \dots + X_n) = \sum_{i=1}^n V(X_i)$.

ex 8: n independent coin flips with bias p .

$X = \# \text{ of heads}$

Observe: $X = X_1 + X_2 + \dots + X_n$

where $X_i \sim \text{Bernoulli}(q)$ are independent.

$$\mathbb{E} X = nq.$$

$$\begin{aligned} V(X) &= \text{theorem} \quad V(X_1) + V(X_2) + \dots + V(X_n) \\ &= n \cdot q(1-q) \quad \text{by ex 7.} \end{aligned}$$

$$q = 1/2.$$

$$\mathbb{E} X = n/2$$

$$V(X) = n \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{n}{4}.$$

$$\text{std dev}(X) = \sqrt{\frac{n}{4}} = \frac{\sqrt{n}}{2}.$$

$$\therefore \frac{\text{Stdev}(X)}{\mathbb{E}X} = \frac{\sqrt{n/2}}{n/2} = \underline{\underline{\frac{1}{\sqrt{n}}}}.$$

i.e. scale of fluctuations is much smaller
than the mean.